

AISHA HABIB

Computer Vision Researcher | Prospective M.S. Candidate, Computer Science & Software Engineering

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[GitHub](#) | [Website](#) | [LinkedIn](#)

SUMMARY

Computer vision and image-processing researcher with a B.E. in Computer Systems Engineering and hands-on experience across object detection, segmentation, and applied deep learning. Delivered end-to-end computer-vision and image-processing systems as a freelance engineer, maintaining a 5-star rating and 100% completion rate, and supported faculty-led research as a Research Assistant, building datasets and evaluating AI/ML models. Led “Smart Acoustical Eye,” a wearable assistive-AI device for visually impaired users that combined real-time object detection with audio-guided environmental awareness, securing competitive national research funding from IGNITE (NGIRI 2022–23) as Project Lead. Co-author on two papers currently under review alongside collaborators already working with Prof. Ping Ping’s group at Hohai University. Seeking to extend this foundation in visual recognition toward an M.S. focused on robust, content-aware image processing and image security.

RESEARCH INTERESTS

Computer Vision and Object Detection, Image Processing and Segmentation, Applied Machine Learning and Deep Learning, Generative Models, Image Content Security and Watermarking.

EDUCATION

B.E., Computer Systems Engineering — MUET

Oct 2019 – Dec 2023

Mehran University of Engineering & Technology (MUET) • Jamshoro, Pakistan • CGPA: 3.51 / 4.0 (Grade A) • FYP: HapticCare — VR-based haptic-jacket therapy system

PUBLICATIONS

Under review

- Imran Ali Soomro, S. Abbas, S. Mazhar, **A. Habib**, T. A. Arain, A. Rahim, “SAG-SSNet: Scene-Adaptive Gated Spatiotemporal Segmentation Network for Robust Winter Road Segmentation,” *Image and Vision Computing* (Elsevier), 2026.
- Imran Ali Soomro, **A. Habib**, S. Abbas, M. H. Zahoor, “A Residual Frequency-Temporal Network for Long-Horizon CPU Utilisation Forecasting in Cloud Data Centres,” *Computing* (Springer).

RESEARCH & PROFESSIONAL EXPERIENCE

Freelance Computer Vision Engineer — Fiverr

Feb 2024 – Present

Hyderabad, Pakistan (Remote)

- Delivered client-specific computer-vision and image-processing projects, maintaining a 5-star rating and 100% completion rate with zero cancellations.
- Built and delivered custom object detection, image classification, and visual-recognition systems end-to-end, from requirement gathering to final delivery.

Research Assistant — Mehran University of Engineering & Technology

Dec 2023 – Jun 2024

Jamshoro, Pakistan • Department of Computer Systems Engineering

- Conducted literature reviews and identified research gaps to support faculty-led research projects.
- Developed and preprocessed datasets for training and testing deep-learning models.
- Designed and evaluated AI/ML models, performing experiments to validate research hypotheses.

SELECTED PROJECTS

Smart Acoustical Eye — wearable assistive-AI device for the visually impaired (Dec 2022 – Nov 2023 • IGNITE NGIRI-funded, Project Lead)

- Engineered a wearable assistive AI device combining real-time object detection with audio-guided environmental awareness in both lit and dark conditions.
- Built an integrated OCR and offline text-to-speech pipeline enabling users to read and understand surrounding text with no internet connectivity.

- Secured competitive national research funding from IGNITE (NGIRI 2022–23) as Project Lead; delivered a fully functional prototype demonstrating real-time multi-object detection with sub-second response time.

FootballVision — *automated football-match analysis pipeline*

- Custom-trained YOLOv5 for player / ball / referee detection; per-player tracking with K-Means team assignment, optical-flow camera-motion compensation, and possession tracking; CLI + Streamlit app.

HaptiCare — Virtual Reality-based system with a haptic jacket for rotator-cuff shoulder therapy.

Snow-Road Slipperiness Detection — Real-time YOLOv8 segmentation of snow-covered roads with slippery-level classification.

Driver Drowsiness Detection — Real-time computer-vision alertness monitor using a deep-learning model deployed on a Raspberry Pi.

TECHNICAL SKILLS

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn; generative models, prompt engineering, transfer learning

Computer Vision: OpenCV, YOLO object detection, MediaPipe; detection, classification, segmentation, visual recognition

NLP / GenAI: NLP, Hugging Face, prompt engineering, generative models

Data / Scientific: NumPy, Pandas, SciPy, Matplotlib, Seaborn, MATLAB

Engineering: Python, Git, Jupyter Notebook, Google Colab, VS Code; data structures & algorithms (arrays, linked lists, trees, sorting & searching); Microsoft Office (Word, Excel, PowerPoint)

HONOURS & AWARDS

IGNITE (NGIRI 2022–2023) FYP Research Funding Grant (2024); Higher Education Commission – PM Youth Laptop Scheme (2023); BISEH, Government of Sindh – Academic Excellence Scholarship for A+ Students.

CERTIFICATIONS & LANGUAGES

Certifications: Advanced Python Programming (NAV TTC Pakistan); Python for Data Science & Machine Learning Bootcamp (Udemy); Applied AI (IBM, Coursera); Python Crash Course (Google); Python Programming & Data Science (Jawan Pakistan)

Languages: Urdu (native); English (Proficient user – CEFR C1–C2)

REFERENCES

Dr. Sanam Narejo — Chairperson, MUET (instructor – Data Science & Software Engineering) • sanam.narejo@faculty.muett.edu.pk

Dr. Shahnawaz Talpur — Associate Professor, MUET (instructor – Web Engineering & Applications, Computer Architecture) •

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